

FSWD Lab

Assignment-4

Objective- To develop Real World Animation Effects such as

- Beating Heart
- Loading Bar/Spinner
- Bouncing Ball
- Flashing Button

Code:-

Beating Heart Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <div class="heart">
      ❤️
    </div>
  </div>
</body>
```

```
</html>
```

Style.css

```
.container{
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}
.heart{
  width: 100px;
  height: 100px;
  font-size: 80px;
  position: relative;
  animation: heartbeat 2s infinite;
```

```
}
```

```
@keyframes heartbeat{  
  0%{  
    transform: scale(1);  
  }  
  15%{  
    transform: scale(2);
```

```
}
```

```
  30%{  
    transform: scale(1);  
  }  
  45%{  
    transform: scale(2);  
  }  
}
```

Loading Bar and Spinner

Index.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport"  
content="width=device-width, initial-  
scale=1.0">  
  <title>Document</title>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>
```

```
<div class="container">  
  <div class="ParentBar">  
    <div class="ChildBar">  
Loading...</div>  
  </div>  
  <div class="ParentSpinner">  
    <div class="ChildSpinner"></div>  
  </div>  
</body>  
</html>
```

Style.css

```
.container{  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  height: 100vh;
```

```
}
```

```
.ParentBar{  
  height:30px;  
  width: 300px;  
  background-color: #ddd;
```

```

    position: relative;
    margin-right: 70px;
}

.ChildBar{
    height: 100%;
    width: 0;
    background-color: lightseagreen;
    animation: loading 2s linear infinite;
    position: absolute;
}

.ParentSpinner{
    height: 60px;
    width: 60px;
    border: 5px solid lightgrey;
    border-top: 5px solid blueviolet;
    border-radius: 50%;
    animation: spin 1s linear infinite;
    position: relative;
}

@keyframes spin{
    0%{
        transform: rotate(0deg);
    }
    100%{
        transform: rotate(360deg);
    }
}

@keyframes loading{
    0%{
        width: 0;
    }
    100%{
        width: 100%;
    }
}

```

Bouncing Ball

Index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
    <title>Document</title>

```

```

    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <div class="ball"></div>
    </div>
</body>
</html>

```

Style.css

```
.container{
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
}
.ball{
  height:100px;
  width:100px;
  background-color:lightcoral;
  border-radius: 50%;
  box-shadow: 10px 10px 10px gray;
```

```
    animation: bounce 2s infinite;
  }

  @keyframes bounce{
    0%{
      transform:translateY(0);
    }
    50%{
      transform: translateY(200px) scaleX(1.1);
    }
  }
}
```

Flashing Button

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Document</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <button class="flash-
button">Emergency</button>
```

```
</div>
</body>
</html>
```

Style.css

```
.container{
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  background-color: black;
}

.flash-button{
  height: 100px;
```

```
width: 200px;  
border-radius: 10px;  
padding: 15px;  
font-size: 20px;  
color: white;  
background-color: crimson;  
animation:flash 2s infinite;  
}
```

```
@keyframes flash{  
  0%{  
    opacity: 1;  
  }  
  50%{  
    opacity: 0.5;  
  }  
}
```

OUTPUT:-



Loading...



Loading...



Emergency

Emergency

Important Notes

1. Top is used to Move an element vertical relative to its position reference point,
2. it works only when position is set for relative absolute fixed or sticky
It does not work with static
3. Task: what if the position is absolute or relative

Ans- if the position of an element is set to relative then the top properly works, The element is moved relative to its original position in the normal document flow, even after moving the original space of the element is still preserved so other elements on the page do not move to fill the gap

4. Task: difference between top and transform and translate in a bouncing ball

Ans- top moves the element by changing its actual position in the layout ,it depend on the position property and can cause layout changes

Transform , translate moves the element visually only without changing its layout position, it is smoother and better for animation like a bouncing ball

In bouncing ball animation translate is preferred because it gives smooth motion and does not disturb other elements

Unique Innovation

Happy Birthday Card

Index.html

```
<!DOCTYPE html>
```

```
<meta charset="UTF-8">
```

```
<html lang="en">
```

```
<head>
```

```

<meta name="viewport"
content="width=device-width, initial-
scale=1.0">

<title>Birthday Card</title>

<link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <div class="Card">
      <h2> 🎉 Happy Birthday! 🎉 </h2>

      <div class="ParentBar">
        <div class="ChildBar">Wishing You
Happiness...</div>
      </div>

      <div class="balloon"> ❤️ 🎈 ❤️ </div>

      <button class="giftBtn">Open Surprise
🎁 </button>
    </div>
  </div>
</body>
</html>

```

Style.css

```

body{
  margin:0;
  font-family: sans-serif;

  background: linear-gradient(135deg, #ff9a9e,
#fad0c4);
}

```

```

.container{
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}

.Card{
  background:white;
  padding:40px;
  width:350px;
  text-align:center;
  border-radius:15px;
  box-shadow:0 10px 25px rgba(0,0,0,0.2);
  border: 10px solid blueviolet;
  position:relative;
  animation:bordermove 2s infinite;
}

h2{
  animation: float 2s infinite ease-in-out;
}

```

```

@keyframes bordermove{
  0%{
    border-color: blueviolet;
  }
  50%{
    border-color: lightseagreen;
  }
}

```



```

    100%{
        border-color: blueviolet;
    }
}

```

```

.ParentBar{
    height:25px;
    width:100%;
    background:#eee;
    margin:20px 0;
    position:relative;
    overflow:hidden;
    border-radius:20px;
}

```

```

.ChildBar{
    height:100%;
    width:0;

    background:linear-gradient(to right,
lightseagreen, teal);
    animation: loading 4s infinite;
    display:flex;
    align-items:center;
    justify-content:flex-end;
    padding-right:10px;
    color:white;
    font-size:12px;
    overflow:hidden;
}

```

```

@keyframes loading{
    0%{width:0;}
    50%{width:100%;}
    100%{width:0;}
}

```

```

/* 🎈 Balloon Floating + Bounce */
.balloon{
    font-size:60px;
    margin:20px 0;
    animation: float 2s infinite ease-in-out;
}

```

```

@keyframes float{
    0%{
        transform: translateY(0) scaleX(1);
    }
    50%{
        transform: translateY(-20px) scaleX(1.1);
    }
    100%{
        transform: translateY(0) scaleX(1);
    }
}

```

```

/* 💎 Flashing Button */
.giftBtn{
    padding:10px 20px;
    border:none;
    border-radius:20px;
}

```

```
background:#ff4e50;
color:white;
font-weight:bold;
cursor:pointer;
animation: flash 1s infinite alternate;
}

@keyframes flash{
  0%{opacity:1;}
  100%{opacity:0.5;}
}
```

